

TS-DP: An Efficient Data Processing Algorithm for Distribution Digital Twin Grid for Industry 5.0

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Abstract—As known, the smart Grid is an essential scenario for Industry 5.0. With its rapid development, the huge number of sensors and smart devices widely used in the industrial field generate significant amounts of data that sharply increase. Facing the power grid environment with a high amount of data, it is easy to cause abnormal conditions in the power grid system and even cause the system to collapse abruptly. To tackle this problem, we simulate a layered digital twin power grid model, using the comparative learning method to process the time series in the power grid and select the positive and negative time series samples using a new slider. In the extended Convolution Neural Network (CNN), we use the Atrous Convolution model, limiting the receptive field and concentrating more data close to the dimension of the model. The time series and process classification tasks are predicted on Electricity Transformer Temperature (ETT), electricity, and University of California, Riverside (UCR) data sets, and experimental results show that the proposed method reduces the error rate of 4.38% and 5.21% in the prediction task and improves the accuracy of 8.39% and 12.73% in the classification task, indications to control the power grid model more accurately, predicting the power grid operation in the future, and take corresponding measures in time.

Index Terms—Contrastive learning, digital twin (DT), industry 5.0, smart grid, time series.

I. INTRODUCTION

WITH the growing demand for renewable energy, the smart grid is a crucial application of Industry 5.0 [1], [2], effectively integrating power system resources and communication infrastructure resources to meet the power consumption needs of users during different periods, as shown in Figure 1 the power Internet of Things (IoT) architecture diagram, with the efficient operation of energy and market and the efficient use of system resources. The power IoT architecture mainly includes the perception layer, network

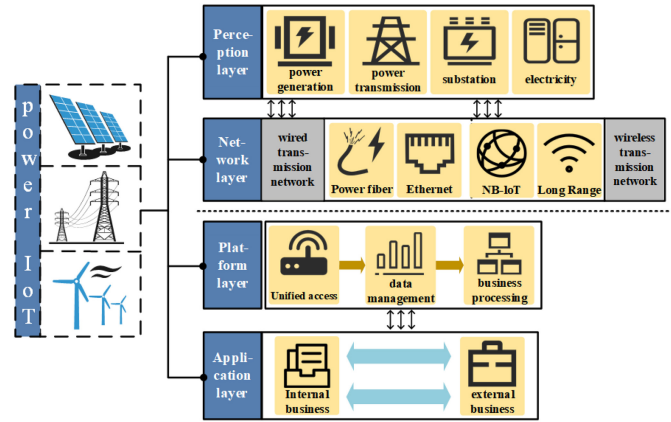


Fig. 1. Grid architecture diagram.

layer, platform layer, and application layer. With the rapid development of power IoT technology, the number of sensors and nodes in it has increased rapidly, whilst the amount of data has risen sharply, which is prone to problems such as data leakage and excessive data load. In recent years, there have been many cases at home and abroad. For example, in March 2019, hackers exploited a known vulnerability in Cisco's firewall to launch a Denial of Service (DoS) attack against a renewable energy utility in Utah, USA, forcing the device to reboot multiple times, causing the group's control center and The communication between the field devices at its various sites is interrupted [3]. In June 2020, hackers extorted 14 million from Brazil-based electricity company Light SA to redeem the use of servers and the hacker's ransomware Sodinokibi ransomware. At the same time, the researchers (also found that the software can enhance privileges by taking advantage of the 32-bit and 64-bit vulnerabilities of cve-2018-8453 vulnerability in Windows win32k components. The existence of this vulnerability will pose a huge security risk to the power IoT [4]. The existing physical power grid model cannot predict the situation in the future and take corresponding measures.

Therefore, it is urgent to adopt a model that combines physical and virtual power grids, and the digital twin distribution network model is suitable for this scenario [5]. As a multi-dimensional and multi-attribute simulation technology, digital twin technology uses sensor data, network communication architecture, entity attributes, and other characteristics to model the actions and states of physical entities and integrates physical entity models and twins through interactive

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channels. The model is mapped and connected in real-time, and the physical system interacts with the virtual space in real-time to form a closed-loop [6]. The digital twin power grid model can predict the system state in the future, simulate the countermeasures to be taken on the virtual power grid, and feedback to the physical power grid according to the execution results of the virtual power grid so that the hidden dangers such as data leakage and overloading of data in the system are discovered as soon as possible, the resulting property losses and personnel losses are minimized, and the safety, reliability, power quality, and transmission efficiency of the power grid system are effectively guaranteed [7].

There are many sensors in the power IoT, and the data of each interrelated sensor affect each other. When a sensor is attacked or the sensor crashes, the data of the sensor will not be available until the sensor returns to normal. How to accurately predict the data of single or multiple sensors based on the connections and characteristics of each interrelated sensor has become a challenge in the data management of the power IoT.

We simulate a digital twin power grid model to solve the abovementioned problems, including hierarchical physical distribution network entities, a data interaction layer, and a digital twin power distribution network. In the model, we use the contrastive learning method. We propose a new data perturbation method to select positive and negative samples. The obtained hidden layer vector is sent to the decoder for time series reconstruction through the input projection layer and timestamp masking. We performed UniVariate and MultiVariate time series prediction and classification tasks on multiple datasets, and this paper has high time series prediction accuracy and classification accuracy.

The remainder of this work is organized as follows: Section II introduces the research status of the digital twin power grid and sensor data prediction technology, Section III introduces the digital twin power grid architecture adopted, and Section IV shows the sensor data prediction algorithm. Experimental results and data analysis are depicted in Section V, and finally, the full text is summarized in Section VI, including directions for future work.

II. RELATED WORK

As an essential technology to promote the social economy, the power industry has gradually become the focus of academia on the information processing method in the smart grid. During the operation of the power grid, data analysis and processing can effectively improve the data support for power dispatching, but the existing methods cannot predict the data of power grid sensors well in the future. For such, we propose in this work an efficient grid sensor data prediction method.

A. Digital Twin Power Grid

The Digital Twin (DT) Power Grid is the mirror image of the physical power grid system based on sensors in the digital space. Jia et al. [8] introduced the concept of digital twins and discussed the construction and application of digital twin technology in the power grid. Yz and Wang [9] introduced the research and application of digital twin technology in power

grid development, realizing the “digital twin” of “physical power grid” and “online power grid”, supporting the high-quality development of power grid business. Pan et al. [10] combined digital twin technology with power systems and applied it to other services such as grid optimization design, grid fault simulation and simulation, virtual power plants, and smart device monitoring. Danileczyk et al. [11] proposed a framework for adapting digital twin technology to microgrid security applications, in which digital twin technology is able to update and improve simulation using real-time bidirectional coupling between simulation and physical systems, resulting in more Well applied to physical systems. Ma et al. [12] realized the interaction between the physical engineering site and the virtual engineering model based on the digital twin technology, improving planning accuracy and coordination. In system infrastructure and IoT applications, Borth et al. [13] discusses the challenges of many tasks related to digital twins and the strategies and architectures for these tasks. From the perspectives of standardized description, knowledge reuse, and data integration, Jiang et al. [5] proposed a digital twin (DTB)-oriented digital portal (DP) model, which supports the construction of DTBs from equipment units to complex systems. Ahmadi et al. [14] proposed a scheme to use digital twin technology at the equipment and system level of electrified railway power systems (ERPS). By analyzing the security risks and characteristics of edge computing, Sun et al. [15] constructed a power grid digital twin security control model including an application layer, function layer, model layer, data layer, and physical layer based on the security protection architecture of edge computing, so as to Make sure all aspects of the grid are under control. Aiming at problems such as power outages, damage to sensitive equipment, and power grids that cannot maintain normal operation, Sleiti et al. [16] proposed a method using deep learning algorithms and a vector autoregressive model to predict the dynamic system model of digital twin technology, which effectively solves the problem. Anomaly detection for power plant data.

B. Sensor Data Prediction Method

The wireless sensor layout is a key link in the construction of the energy Internet and an important support for the power grid’s digital, networked, and intelligent development. During the power grid’s operation, the power data analysis and prediction improve the smart grid’s effective utilization. Kumar Gilbert et al. [17] proposed a new method for security data prediction in wireless sensor networks, which uses a Toeplitz matrix-based time series trust model (TSTM) and a trust-based autoregressive (TAR) process to Better performance in data prediction and Compressed Sensing (CS) based aggregation and reconstruction. Reference [18] proposed a deep long short-term memory (DLSTM) network-based RUL prediction method using multiple sensor time-series signals. The model fuses multi-sensor monitoring signals for accurate RUL prediction, which can discover hidden long-term dependencies between sensor time-series signals through a deep learning structure, improving data prediction accuracy. Reference [19] developed a single-stage detector and

predictor that utilizes 3D point clouds produced by LiDAR sensors and a dynamic map of the environment, achieving better accuracy and saving computation. To extract generalized features and latent relationships in power load-related edge-sensing data, Liu et al. [20] proposed a power-load forecasting scheme based on edge-sensing data-imaging conversion (DIC) to improve the smart city and social forecasting precision. Darvishi et al. [21] proposed a real-time generalized machine learning architecture for sensor-based validation based on a series of neural network estimators and classifiers. Ren and Cao [22] proposed a method to process Computational Fluid Dynamics data using the LLVM method to generate further a low-dimensional database for ANN predictions using monitored concentrations from different sensor layouts (i.e., location and number) for a series of ANN predicts the input and improves the accuracy of data prediction. Aiming at the problems of multi-party sensor data sharing and privacy protection, and realizing cross-organizational traffic data fusion and prediction, Wang et al. [23] proposed an accurate traffic prediction method based on LSH (locality-sensitive hashing). Feasibility of prediction accuracy and efficiency while maintaining the privacy of sensor data. Balogun et al. [24] preprocessed some data from IoT sensors, such as weather and traffic data corresponding to time, and used these data to develop a machine learning prediction model for NO₂ pollution concentration. Bodenstedt and Wagner [25] proposed a context-aware method based on a convolutional neural network that can analyze the intervention workflow online and automatically predict the remaining duration, improving the prediction accuracy of online surgery duration. Yang et al. [26] proposed a digital twin-driven composite fault diagnosis method by combining virtual and real data, which is extremely effective for composite faults in subsea production control systems.

C. Sensor Data Synchronization Method

Sensor networks perform complex monitoring and sensing tasks collaboratively through various integrated micro-sensors. However, due to the unpredictability and imperfect measurability of message delays in the IoT environment, the data synchronization problem of sensor nodes has caused researchers and scholars' attention. Sguazza et al. [27] proposed an offline data-driven synchronization solution, which can process data of different nature and sample at different frequencies, which not only solves the problem of data synchronization but also solves the problem of clock drift caused by Data time alignment problem. Since the clock will be skewed, Skiadopoulos et al. [28] propose a lightweight synchronization algorithm for wireless sensors, which solves the problem of time skew and corresponding variance in harsh environments such as IoT systems. Due to the local clock synchronization problem of wireless sensor network nodes, Chalapathi et al. [29] proposed a time synchronization protocol called Efficient Simple Time Synchronization Algorithm (E-SATS) for cluster-based wireless sensor networks. Volume and energy consumption are significantly reduced. Xu et al. [30] designed a data set synchronization protocol for wireless

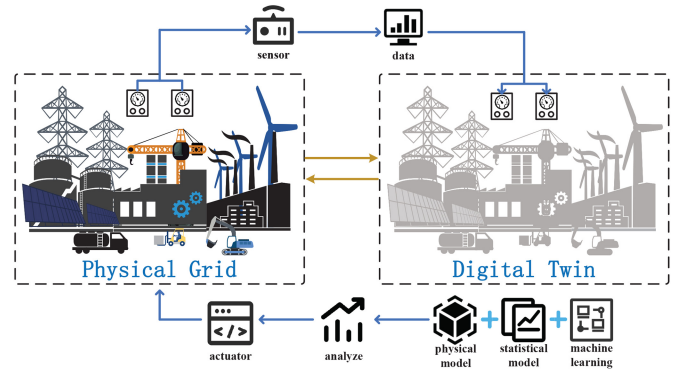


Fig. 2. Digital twin grid model.

sensor networks, which uses a sleep mechanism to save energy and optimize energy consumption. Since data and energy consumption can be minimized in wireless sensor networks, Palanisamy et al. [31] proposed a Multi-Sensor Data Synchronization Scheduling (MSDSS) framework for efficient scheduling at the receivers of heterogeneous sensor networks data aggregation. Due to the lack of spatial and temporal synchronization of roadside sensors due to different in-vehicle devices, Du et al. [32] proposed a novel spatiotemporal synchronization method for asynchronous roadside mmWave radar cameras for sensor fusion, which utilizes scene. The feature extraction of lane line corners to pre-calibrate the camera solves the problem of spatiotemporal synchronization of roadside sensors. Aiming at the dataset analysis problem of multiple sensor systems in synchronizing the Internet of Vehicles, Ishiwatari et al. [33] proposed a method for synchronizing video data with acceleration data from on-board sensors of a moving vehicle, which uses image features to detect synchronization point and then match it with the corresponding point in the acceleration data. Since the data of IoT devices are all sent to cloud storage, Wang et al. [34] proposed a new architecture for data synchronization based on fog computing, aiming at the privacy and security issues that may arise in this process. Offloading to the fog server can ensure data privacy and security.

III. GRID DIGITAL TWIN

This section constructs the digital twin distribution network model, as shown in Figure 2. Firstly, a hierarchical distribution network entity is established, including distribution network distributed generation, static var compensator, energy storage model, and distribution network hierarchical model, which provides real-time data for the data interaction layer and completes the dynamic adjustment of the relationship between physical entities. Secondly, the data interaction layer is constructed, which is mainly divided into a data transmission module and a data perception module. According to the data interaction layer mapping, the distribution network entity model is transmitted to the digital twin, the digital twin model is constructed, the data of the physical distribution network is fully explored and extracted, the data characteristics of the power system are extracted, and the results processed in the virtual distribution network are fed back to the physical

distribution network through the data interaction layer. The physical distribution network executes better strategies according to the results.

A. Hierarchical Distribution Grid Entities

The power transmission network or power plant outputs power to the distribution network, and the distribution network distributes electricity energy to various users utilizing step-by-step distribution. The distribution network is mainly composed of equipment such as cables, towers, overhead lines, power sources [35], reactive power compensators [36], and energy storage models [37], combined with a hierarchical topology model [38]. Model key equipment in a hierarchical distribution network for digital twinning of the physical entities of the distribution network. Provide real-time real data for the data interaction layer, record the event occurrence process, complete the dynamic adjustment of the relationship between physical entities, and execute the latest feedback instructions in the digital twin through the data interaction layer.

1) *Distributed Power Supply in the Distribution Network*: In this module, we use the micro-turbine as a representative to analyze the characteristics of distributed power in the distribution network. The Micro Gas Turbine (MGT) is connected to the distribution network, and the active and reactive power output can be adjusted independently using the inverter multiplexing method. Among them, the inverter capacity still limits the active and reactive power output of MTG. $P_{\max}(MGT)$ and $P_{\min}(MGT)$ are the upper and lower limits of the MGT active power output, $S(MGT)$ is the inverter capacity, $P_{\max}(MGT)$ and $P_{\min}(MGT)$ are the upper and lower limits of the MGT ramp rate, and $P_{j,t}(MGT)$ is the active power output of the j th MGT in the t period, and $Q_{j,t}(MGT)$ is the reactive power of the j th MGT in the t period. The MGT model is expressed as follows:

$$P_{\min}(MGT) \leq P_{j,t}(MGT) \leq P_{\max}(MGT) \quad (1)$$

$$p_{\min}(MGT) \leq P_{j,t}(MGT) - P_{j,t-1}(MGT) \leq p_{\max}(MGT) \quad (2)$$

$$P_{j,t}^2(MGT) + Q_{j,t}^2(MGT) \leq S^2(MGT) \quad (3)$$

2) *Static Var Compensator (SVC)*: At present, the SVC method with fast response speed and high stability is mainly the TSC (Thyristor Switched Capacitor) + TCR (Thyristor Controlled Reactor) model, through which the switching of groups can be realized, and the TCR method can be used for branch coordination. And can carry out continuous adjustments of reactive power. Among them, $\alpha(90^\circ \leq \alpha \leq 180^\circ)$ is the trigger delay angle of the thyristor, U is the voltage of the connection point of the SVC, B_C is the fundamental susceptance of the TSC, and $B_L(\alpha)$ is the equivalent fundamental susceptance of the TCR.

$$B_C = \omega C \quad (4)$$

$$B_L(\alpha) = \frac{2(\pi - \alpha) + \sin 2\alpha}{\pi \omega L} \quad (5)$$

$$B(SVC) = \frac{\pi \omega^2 LC - 2(\pi - \alpha) - \sin 2\alpha}{\pi \omega L} \quad (6)$$

$$Q(SVC) = [B_C - B_L(\alpha)]U^2 = B_{SVC}^2 U \quad (7)$$

SVC uses thyristors to control capacitors, reactance elements, and other equipments, mainly by adjusting the TSG situation, changing the reactive power data Q SVC of the SVC system, and finally controlling the bus voltage within a reasonable range, and continuously adjusting.

3) *Energy Storage Model*: The energy storage model set in this paper fully considers the individual capacity, state of charge (SOC), power and other factors, the energy storage individual q , the rated capacity $Content_q$ of the individual q , and $Content_{\sum_q}$ is the total amount of energy storage individuals in the system. The adaptive filling function of the input (output) power of the energy storage individual q is as follows:

$$p(s_q) = \frac{Content_q P_{q, rat} A_q e^{d_q f(s_q)}}{Content_{\sum_q} \times (P_{q, rat} + A_{q0} e^{d_q f(s_q)})} \quad (8)$$

Among them, $P_{q, rat}$ is the rated power of the individual energy storage, A_q is the adaptive factor of the individual energy storage, and d_q is the automatic distribution coefficient of the individual. According to the state of charge and state of discharge, the adaptive charge-discharge function of the energy storage individual q is selected.

$$p(s_q)_{charge} = Content_q P_{q, rat} A_q \exp\left(\frac{d_q [(s_{shc} + s_{min})/2 - P_q]}{s_{min} - s_{min}}\right) / \left(Content_{\sum_q} \times \left(P_{q, rat} + A_{q0} \exp\left(\frac{d_q [-(s_{shc} + s_{min})/2 - P_q]}{s_{min} - s_{min}}\right) \right) \right) \quad (9)$$

$$p(s_q)_{discharge} = Content_q P_{q, rat} A_q \exp\left(\frac{d_q [s_q - (s_{mid} + s_{shd})/2]}{s_{min} - s_{min}}\right) / \left(Content_{\sum_q} \times \left(P_{q, rat} + A_{q0} \exp\left(\frac{d_q [s_q - (s_{mid} + s_{shd})/2]}{s_{max} - s_{mid}}\right) \right) \right) \quad (10)$$

Among them, $p(s_q)_{charge}$ and $p(s_q)_{discharge}$ are the charge-discharge functions in the charge and discharge states, respectively.

4) *Layered Model of Distribution Network*: For nodes in the distribution network, we use the adjacency matrix to represent. The layered model of the distribution network takes a node as the core node, is layered according to the path length between other nodes and this node, and defines the layered identification matrix of the distribution network. The identification matrix is as follows:

$$iden_{ij} = \begin{cases} g_{ij}^l, g_{ij}^l = 1, i \neq j, l = 1, \dots, N-1 \\ 0, g_{ij}^l \neq 1, i \neq j, l = 1, \dots, N-1 \\ 0, i = j, l = 1, \dots, N-1 \end{cases} \quad (11)$$

where g_{ij}^l represents the path length l from node i to node j

$$\begin{cases} Layer_1, d_{ij} = (d_{1,1}, d_{1,2}, \dots, d_{1,N}) \\ Layer_2, d_{ij} = (d_{2,1}, d_{2,2}, \dots, d_{2,N}) \\ \vdots \\ Layer_N, d_{ij} = (d_{N,1}, d_{N,2}, \dots, d_{N,N}) \end{cases} \quad (12)$$

B. Data Interaction Layer

The data interaction layer is mainly divided into a data transmission module [39] and a data perception module. As

the data transmission channel of the physical entities and twins of the distribution network, the data transmission module can effectively acquire, transmit, and coordinate information. The data perception module can sense the data of sensors, collectors, meters, and other equipment according to different power grid scenarios and needs and share the resources of the whole system through the collected data. The main data sensing operations include: installing signal collectors on transmission equipment, installing protection measurement and control devices on power equipment at all levels, and installing sensors on terminal ring network cabinet equipment.

The physical distribution network entity has incoming data packets in the finite domain Ω , and the data transmission model fuses each broadcast data packet into a linear representation of the original data packet. $M = \{m_1, m_2, \dots, m_n\}$ represents the set of n original data packets, and D_i represents the linear combination of all data packets:

$$D_i = \sum_{j=1}^n m_j \rho_{j,i} \quad (13)$$

Among them, $\rho_{j,i}$ represents the random network coding coefficient of the data packet sent by node J in Ω . Extracting the encoding vector $\{K_1, K_2, \dots, K_n\}$ from the encoded message D_i , a node with M encoded messages $e_1, e_2, \dots, e_M$ solves the system of linear equations, which can be expressed as:

$$M_{t_{M \times 1}} = Q_{M \times n} \chi_{n \times 1} \quad (14)$$

Among them, all embedded coding vectors form a matrix $Q_{M \times n}$. If the rank of the Q matrix is not less than n , it means that the digital twin model successfully receives the data transmitted by the physical entity of the distribution network.

According to the real environment and usage requirements of the distribution network, we deploy sensing nodes in important parts of the distribution network, collect sensor data that may affect the normal operation of the distribution network, and build a relationship database between the operating environment of the distribution network and the status of key sensors. We use the data perception method based on Moving Variance to compare the data of the same node in different time windows through variance. When the comparison result exceeds the set threshold, it means that the data fluctuates greatly in the two-time windows. The data-aware approach can be expressed as:

$$TD = t_e - t_s \quad (15)$$

$$\bar{S}_t = \frac{\sum_{i=k}^{k+win} S_i}{win} \quad (16)$$

$$Var_t = \frac{\sum_{i=k}^{k+win-1} (S_i - \bar{S}_t)^2}{win - 1} \quad (17)$$

$$\overline{Var}_t = \frac{\sum_{i=t_s}^{t_e} Var_i}{TD + 1} \quad (18)$$

$$\sigma_{win} = \frac{\sum_{i=t_s}^{t_e} (Var_i - \overline{Var}_t)^2}{win - 1} \quad (19)$$

Among them, win represents the size of the time window used, $\bar{S}_t$ is the average value of the data in the window, $\overline{Var}_t$

represents the average value at a time, and σ_{win} represents the standard deviation at time TD . At the time i , $Var_i > \overline{Var}_t + \theta \sigma_{win}$ indicates that the data fluctuation is greater than the set threshold, and θ indicates the threshold parameter, which needs to be corrected for the data in the TD time.

C. Digital Twin Mapping Networks

According to the data interaction layer mapping, the physical model of the distribution network is transferred to the digital twin, the data of the physical distribution network is fully explored and extracted, and then the data of the physical distribution network is detected, identified, classified, repaired, etc. The operation, processing the related business of the distribution network in the virtual distribution network, extracting the data characteristics of the power system, and feeding back the results processed in the virtual distribution network to the physical distribution network through the data interaction layer, and the physical distribution network according to the results. Implement better strategies.

IV. TS-DP: GENERAL SENSOR DATA PROCESSING ALGORITHMS

This section describes the sensor data processing algorithm in the digital twin grid model for predicting future grid sensor data. The most common method in machine learning is supervised learning. Using the supervised learning method, we can get a set of labeled data (X, Y) , that is, feature-label pairs, and the main task is to learn the relationship between features and label pairs. Model training using the supervised learning method requires running a large amount of data, and the process of labeling data for millions of data will consume a lot of computing power and time. The model trained by the supervised learning method is very good at solving the current task, but because the network of supervised learning only focuses on the good representation of X and the (X, Y) relationship, it cannot represent a general model for learning X , and cannot well generalize the model to related fields.

Self-supervised learning, a subcategory of supervised learning, is also suitable for learning (X, Y) label (X, Y) pairs, but without the need for manual labeling of the dataset. The basic idea of self-supervised learning is to hide a certain part of the input data, and then use the hidden part to observe its value of the hidden part. Using self-supervised learning can lead to a more general range of models that can be fine-tuned for downstream tasks. Self-supervised learning usually requires us to consider the loss function of the supervised process. We do not pay much attention to the final performance of the task. More attention is focused on the intermediate data representations in the learning process. For these representations, we hope to cover more abundant semantics and can improve the performance of downstream tasks.

This paper improves the model depicted in [40], performs perturbation processing on the collected data, and selects positive and negative sample data. Then the original data is sent to the encoder, the input time series is mapped to a high-dimensional latent vector through the input projection layer, and a random timestamp is selected to mask the latent vector

Algorithm 1 Data Perturbation Method With Half of the Positive and Negative Samples

```

1: sections = input_data //Crop training data according to
  preset maximum processing length
2: for i <= section do
3:   if input_date[x,y] == NaN then //If there are empty
    values, fill with the mean of the current column
4:     input_date[x, y] = mean(input_date[y])
5:     train_data=joint_as_dimention_0input_date[section]
    //The input data is cut into equal parts of sections, and
    the data is spliced according to the 0-dimension to
    obtain the training data
6:     crop_random=random(low=0,height=size(train_data))
    //pick any point
7:     crop_left = random(0,random)
8:     crop_right = crop_left + crop_random //middle over-
    lapping box
9:     crop_eleft = max(crop_left - crop_l-2×crop_left,0)
10:    crop_eright=min(2×crop_right+crop_l,size(
    train_data))
11:    out1 = train_data[crop_eleft,crop_right] //get two
    slices
12:    out2 = train_data[crop_left,crop_eright]

```

to generate a context view. We use the representation learning method to use the learned hidden layer vector, use the decoder to map the hidden layer vector back to the input vector, reconstruct, and achieve the purpose of data dimensionality reduction and include more semantic features.

In the data perturbation module, we use the method of two sliders to select positive and negative samples. The two sliders represent two time periods $[x_1, x_2]$ and $[y_1, y_2]$, $0 < x_1 \leq x_2 \leq y_1 \leq y_2 \leq \text{Time}$, and Time represents the length of the time series. We use a new data perturbation method, that is, $[x_1, x_2]$, the length of $[x_1, y_1]$ and $[y_1, y_2]$ are consistent, we take $[x_1, y_1]$ as a positive sample, $[x_1, x_2]$ and $[y_1, y_2]$ as a negative sample, and perform two positive and negative samples in the two time periods of $[y_1, y_2]$ and $[y_1, y_2]$ study comparison. The data perturbation method is shown in Algorithm 1. First, the training data is cropped by the preset maximum processing length. If there is a null value in the input data, the mean value of the column is used to fill in, and then the cropped data is processed, etc. The training data is obtained by splicing according to dimension 0, a period of time sequence is randomly selected from the training data, and the data perturbation method is executed.

In the encoder, we use atrous convolution as an extended CNN, adopting an atrous convolution that is more inclined to the same dimension data, as shown in Figure 3. In the first five layers, the dilation of the Atrous Convolution is continuously increased to enlarge the receptive field. The last five layers of convolution stop expanding and lock the dilation number to 2; that is, in the later stage of feature extraction, we pay more attention to the same dimension as our data reduce the interference of other dimension data on model training. Since we limit the receptive field of Atrous convolution, we have higher accuracy in univariate prediction, and when the

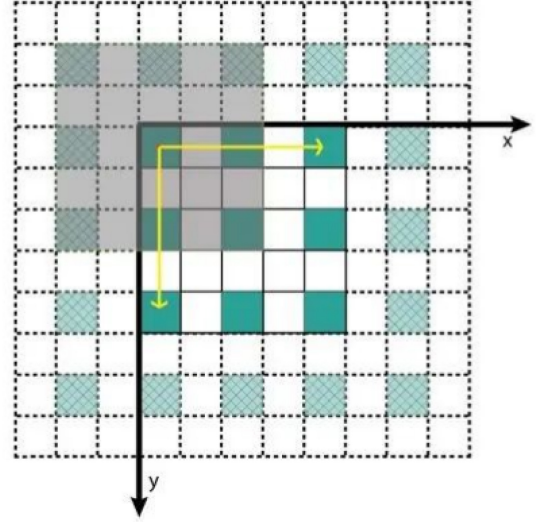


Fig. 3. Atrous convolution.

Algorithm 2 Prefer Atrous Convolution for Data of the Same Dimension

```

1: input_data[] = train_data[]
2: feature_extractor[] = DilatedConvEncoder(
  input_data_dimention, out_channels, kernel_size, dilation)
3: if input_Data_type = multivar
4:   for i <= DilatedConv_deep do
5:     if i > 5
6:       dilation = 2
7:     else
8:       2^i //Dilation=2 for more than 5 layers of Dilated
        convolution
9:   for i <= DilatedConv_deep do
10:    dilation = 2^i
11:    return feature_extractor[] //return feature matrix

```

data dimension is smaller and can achieve better classification accuracy in most datasets with smaller data dimensions, such as Algorithm 2 shown.

In this section, a general sensor data processing algorithm is constructed. We use the contrastive learning method to construct positive and negative samples through a special data perturbation method. The perturbed data is sent to the encoder, and the input projection layer is mapped to high-dimensional data. Then pass in the Atrous Convolution layer. Since the change of the Atrous Convolution is limited, this layer pays more attention to the data of the same dimension, to reduce the interference caused by the data of other dimensions to the model training.

V. EXPERIMENT AND DISCUSSION

All experimentations in this work are performed on a server equipped with one Intel CORE i9-11990K CPU, 64G memory, and one NVIDIA T4 GPU accelerator card, installed with CUDA version 11.3, Python 3.8, and PyTorch 1.8.2, Max threads and batch size configuration set to 8 on each dataset.

TABLE I
RESULTS OF UNIVARIATE TIME SERIES FORECASTING BY DIFFERENT METHODS

		Our Model		TS2Vec		Informer		LogTrans		N-BEATS		TCN		LSTnet	
Dataset	H	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	24	0.047	0.167	0.039	0.152	0.098	0.247	0.103	0.259	0.094	0.238	0.075	0.21	0.108	0.284
	48	0.075	0.21	0.062	0.191	0.158	0.319	0.167	0.328	0.21	0.367	0.227	0.402	0.175	0.424
	168	0.143	0.295	0.134	0.282	0.183	0.346	0.207	0.375	0.232	0.391	0.316	0.493	0.396	0.504
	336	0.16	0.32	0.154	0.31	0.222	0.387	0.23	0.398	0.232	0.388	0.306	0.495	0.468	0.593
ETTh2	24	0.089	0.224	0.09	0.229	0.093	0.24	0.102	0.255	0.198	0.345	0.103	0.249	3.554	0.445
	48	0.126	0.275	0.124	0.273	0.155	0.314	0.169	0.348	0.234	0.386	0.142	0.29	3.19	0.474
	168	0.225	0.379	0.208	0.36	0.232	0.389	0.246	0.422	0.331	0.453	0.227	0.376	2.8	0.595
	336	0.22	0.376	0.213	0.369	0.263	0.417	0.267	0.437	0.431	0.508	0.296	0.43	2.753	0.738
ETTm1	24	0.014	0.089	0.015	0.092	0.03	0.137	0.065	0.202	0.054	0.184	0.041	0.157	0.09	0.206
	48	0.025	0.118	0.027	0.126	0.069	0.203	0.078	0.22	0.19	0.361	0.101	0.257	0.179	0.306
	96	0.043	0.157	0.044	0.161	0.194	0.372	0.199	0.386	0.183	0.353	0.142	0.311	0.272	0.399
	288	0.106	0.25	0.103	0.246	0.401	0.554	0.411	0.572	0.186	0.362	0.318	0.472	0.462	0.558
	672	0.149	0.299	0.156	0.307	0.512	0.644	0.598	0.702	0.197	0.368	0.397	0.547	0.639	0.697
Electricity	24	0.253	0.287	0.26	0.288	0.251	0.275	0.528	0.447	0.427	0.33	0.263	0.279	0.281	0.287
	48	0.3	0.314	0.319	0.324	0.346	0.339	0.409	0.414	0.551	0.392	0.373	0.344	0.381	0.366
	168	0.421	0.387	0.427	0.394	0.544	0.424	0.959	0.612	0.893	0.538	0.609	0.462	0.599	0.5
	336	0.557	0.474	0.565	0.474	0.713	0.512	1.079	0.639	1.035	0.669	0.855	0.606	0.823	0.624
Avg.		0.174	0.272	0.209	0.296	0.310	0.390	0.370	0.434	0.399	0.426	0.338	0.413	1.099	0.536

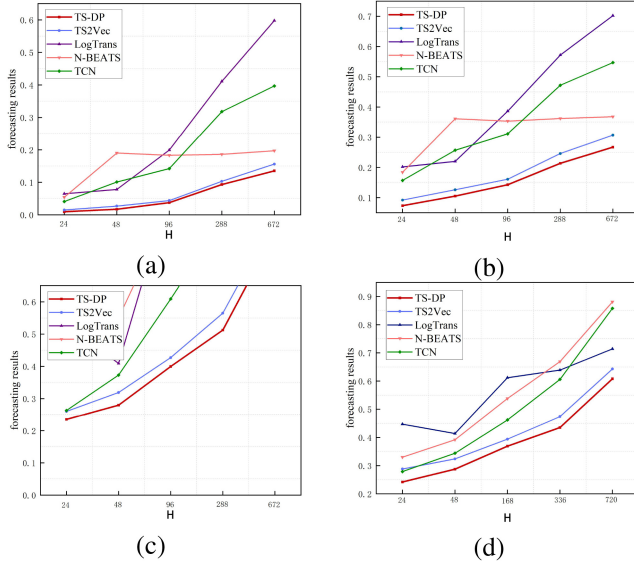


Fig. 4. Results of Univariate predictions performed by multiple methods on different datasets. Caption: Figure 4(a) shows the MSE results of Univariate prediction in ETTh1, Figure 4(b) shows the MAE results of Univariate prediction in ETTh2, Figure 4(c) shows the MSE results of Univariate prediction in ETTm1, Figure 4(d) Represents the MAE result of Univariate prediction in Electric.

The datasets used include:

- ETT data set [41]: divided into multiple sub-datasets ETTh1, ETTh2, and ETTm1, mainly related to parameters such as oil temperature and six power load characteristics, which can be predicted by the relevant data.
- Electricity dataset [42]: is used for research in areas such as electric load forecasting and energy market analysis.

The dataset records electric load data for the power system over a period of time, usually sampled at hourly or 15-minute intervals.

- UCR dataset [43]: is the “Imagnet” of the time series field, containing time series data from many different domains and applications. Each dataset contains a series of time series samples with corresponding labels.

We perform time series prediction tasks in ETT and Electricity datasets and time series classification tasks in UCR datasets.

We use the following methods for comparison in data prediction experiments:

- TS2Vec [40]: is a universal framework for learning representations of time series in an arbitrary semantic level.
- Informer [41]: Beyond Efficient Transformer for Long Sequence Time-Series Forecasting.
- LogTrans [44]: improved the prediction accuracy of time series with limited memory budget.
- N-BEATS [45]: Neural basis expansion analysis for interpretable time series forecasting.
- TCN [46]: is a sequence modeling benchmarks and Temporal Convolutional Networks.
- LSTnet [47]: is a Long- and Short-term Time-series network.
- StemGNN [48]: is a Spectral temporal graph neural network for multivariate time-series forecasting.

The following methods are used for comparison in data classification experiments: TS2Vec [40], T-Loss [49], TNC [50], TSTCC [51], DTW [52], TST [53].

The accuracy of the model is usually measured using mean square error (L2 loss) MSE and mean absolute value error

TABLE II
RESULTS OF MULTIVARIATE TIME SERIES FORECASTING BY DIFFERENT METHODS

Dataset	H	Our Model		TS2Vec		Informer		StemGNN		TCN		LogTrans		LSTNet	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	24	0.554	0.513	0.599	0.534	0.577	0.549	0.614	0.571	0.767	0.612	0.686	0.604	1.293	0.901
	48	0.591	0.54	0.629	0.555	0.685	0.625	0.748	0.618	0.713	0.617	0.766	0.757	1.456	0.96
	168	0.732	0.625	0.755	0.636	0.931	0.752	0.663	0.608	0.995	0.738	1.002	0.846	1.997	1.214
	336	0.912	0.717	0.907	0.717	1.128	0.873	0.927	0.73	1.175	0.8	1.362	0.952	2.655	1.369
ETTh2	24	0.416	0.481	0.398	0.461	0.72	0.665	1.292	0.883	1.365	0.888	0.828	0.75	2.742	1.457
	48	0.574	0.578	0.58	0.573	1.457	1.001	1.099	0.847	1.395	0.96	1.806	1.034	3.567	1.687
	168	1.947	1.081	1.901	1.065	3.489	1.515	2.282	1.228	3.166	1.407	4.07	1.681	3.242	2.513
	336	2.369	1.227	2.304	1.215	2.723	1.34	3.086	1.351	3.256	1.481	3.875	1.763	2.544	2.591
ETTh1	24	0.428	0.428	0.443	0.436	0.323	0.369	0.62	0.57	0.324	0.374	0.419	0.412	1.968	1.17
	48	0.566	0.509	0.582	0.515	0.494	0.503	0.744	0.628	0.477	0.45	0.507	0.583	1.999	1.215
	96	0.618	0.545	0.622	0.549	0.678	0.614	0.709	0.624	0.636	0.602	0.768	0.792	2.762	1.542
	288	0.707	0.602	0.709	0.609	1.056	0.786	0.843	0.683	1.27	1.351	1.462	1.32	1.257	2.076
	672	0.786	0.652	0.786	0.655	1.192	0.926	—	—	1.381	1.467	1.669	1.461	1.917	2.941
Avg.		0.861	0.653	0.862	0.655	1.188	0.809	1.135	0.778	1.301	0.903	1.478	0.996	2.261	1.664

(also known as an L1 loss) MAE. MSE is less sensitive to outliers than MAE and can be used to evaluate the global accuracy of the model, while MAE can better reflect individual outliers. The formula is as follows:

$$\text{MSE} = \frac{1}{m} \sum_{i=1}^m (y_i - \hat{y}_i)^2 \quad (20)$$

$$\text{MAE} = \frac{1}{m} \sum_{i=1}^m |y_i - \hat{y}_i| \quad (21)$$

We employ atrous convolutions that prefer same-dimensional data, as shown in Algorithm 2. In the first five layers, the dilation of the Atrous Convolution is continuously increased to increase the receptive field. The last five layers of convolution stop expanding and lock the dilation number to 2. That is, in the later stage of feature extraction, we pay more attention to the same dimension as ourselves. Data to reduce the interference of other dimension data on model training. As shown in Table I and Table II, we conducted time series prediction experiments on the ETT and the Electricity datasets, then compared them with the current state-of-the-art methods. Since we limit the receptive field of the atrous convolution, the Higher accuracy when making univariate forecasts and when data dimensions are smaller.

Figure 4 and Figure 5 show the Univariate and MultiVariate prediction results of Our Model and TS2Vec, LogTrans, N-BEATS, and TCN methods on the ETTh1, ETTm1, and Electricity datasets. With the increase of H, our method still maintains a low value in terms of MAE and MSE, indicating that the method in this paper has higher accuracy for prediction on ETTh1, ETTm1, and Electricity datasets.

Due to the reduced receptive field, our method can achieve better classification accuracy in most datasets with smaller data dimensions. Figure 6 shows the accurate scores of Ours, TS2Vec, T-Loss, TNC, TS-TCC, TST, and DTW methods for

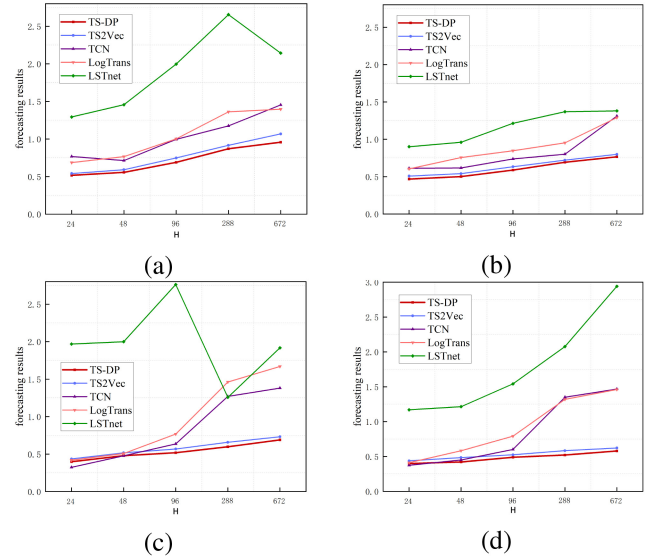


Fig. 5. Results of Multivariate predictions by multiple methods on different datasets. Figure 5(a) shows the MSE results of Multivariate prediction at ETTh1, Figure 5(b) shows the MAE results of Multivariate prediction at ETTh1, Figure 5(c) shows the MSE results of Multivariate prediction at ETTm1, and Figure 5(d) represents the MAE result of Multivariate prediction at ETTm1.

data classification on 10 UCR. Compared with other methods, the method in this paper has the highest score in most UCR datasets except BeetleFly, PLAID, and Car, and the ranking in BeetleFly, PLAID, and Car datasets is not the highest, but it also achieves a relatively high score.

As can be seen from Figure 7, our loss curve has significantly smaller fluctuations than ts2vec, and can also converge quickly. The fluctuation range of the Loss curve is often related to three factors: batch size, learning rate, and whether the data passed to the model for learning is what the model needs. In order to exclude the influence of other factors, we reproduced ts2vec on Nvidia T4 GPU with a consistent batch size set to

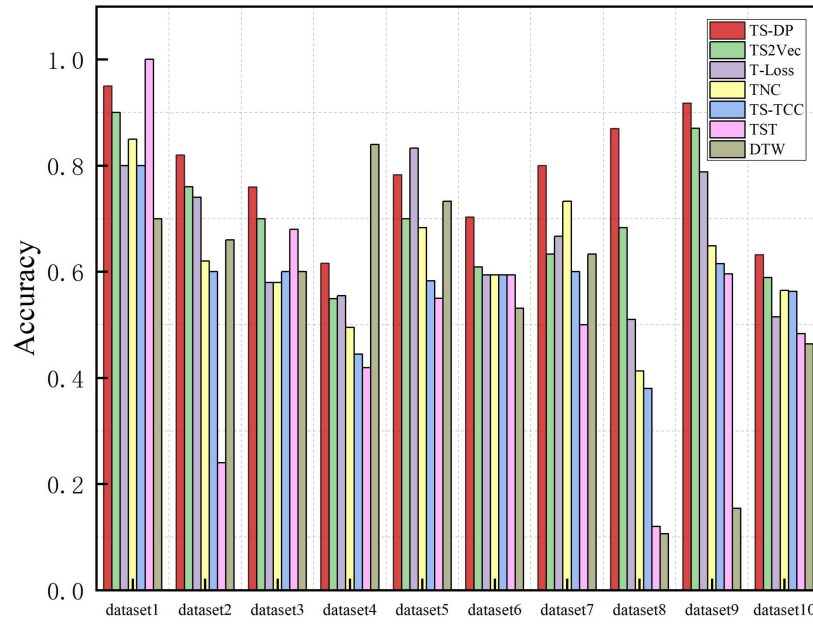


Fig. 6. Accuracy comparison of different methods for data classification on 10 UCR datasets. Caption: dataset1 represents BeetleFly, dataset2 represents PickupGestureWiimoteZ, dataset3 represents Rock, dataset4 represents PLAID, dataset5 represents Car, dataset6 represents Herring, dataset7 represents Beef, dataset8 represents PigAirwayPressure, dataset9 represents PigCVP, and dataset10 represents RefrigerationDevices.

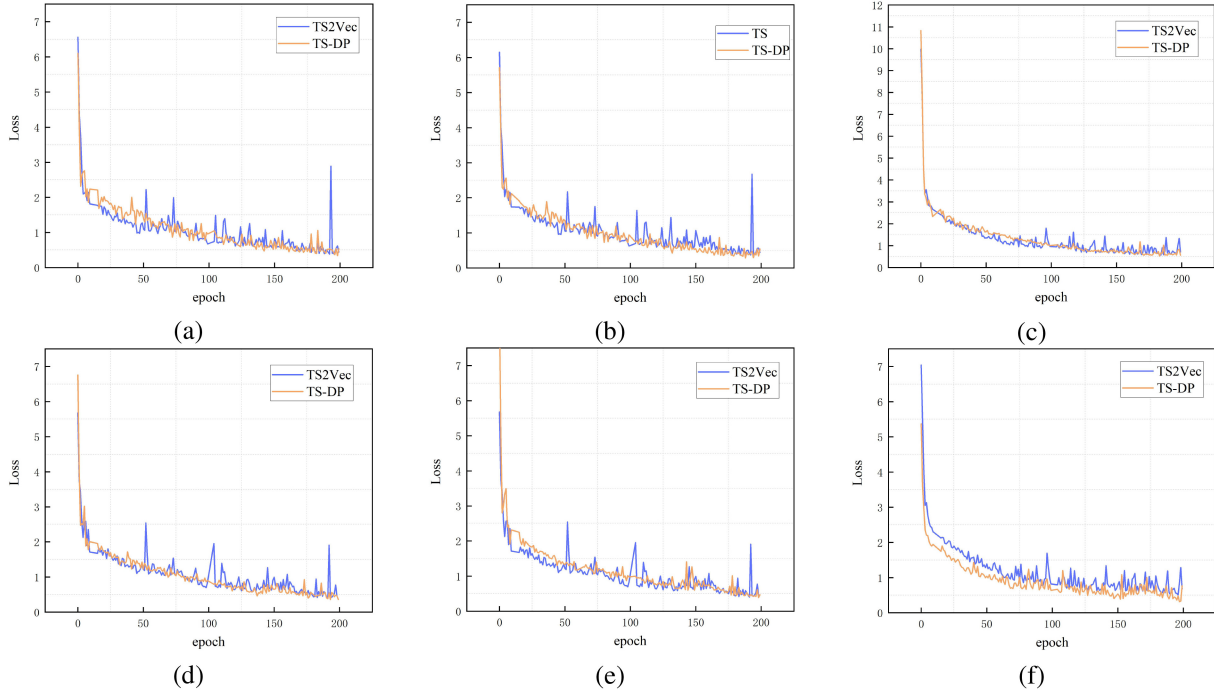


Fig. 7. In this work, the proposed method is compared with the TS2Vec method for the prediction and training of Univariate and Multivariate loss values on different datasets. Figures 7(a), (b), and (c) show the comparison of Loss values for model training on the ETTh1, ETTh2, and ETTm1 datasets, respectively, while Figures 7(d), (e), and (f) show the comparison of Loss values for model training on ETTh1, ETTh2, and ETTm1 datasets, respectively.

8 and a learning rate set to 0.001. Every time we perturb the data, we ensure that the ratio of positive and negative samples is 1 to 1, which makes the model update more necessary when we input the training data so that our loss can be smoothly attenuated with a smaller fluctuation range. This means that our data perturbation method is more robust in the face of more unevenly distributed data.

The experimental results show that the proposed method better affects time series prediction tasks and classification

tasks and can handle time series better. Since we limit the expansion speed of atrous convolution, we have higher accuracy in univariate prediction and when the data dimension is smaller. In the prediction task, our method improves the Univariate prediction of the dataset Electricity by 5.21%. The dataset ETTm1 improved by 4.38% for Multivariate prediction. We performed the time series classification task on the UCR dataset, which improved by 12.73% on the dataset PigAirwayPressure and 8.39% on the dataset Beef.

VI. ENGINEERING APPLICATIONS

Digital Twin (DT) technology is a virtual modeling-based technology that creates a virtual copy of the power system in a computer to simulate and optimize the entire power system. In this work, DT technology is used to simulate the operation of the power system and optimize the stability and safety of a power system. The algorithms proposed in this research have suitable engineering applications for power grid fault diagnosis and prediction, intelligent dispatching, and control, which are detailed next.

- In fault diagnosis and prediction: digital twin technology can identify faults in the power system and predict possible future faults by analyzing and processing the data of the power system. In this investigation, we use digital twin technology to diagnose faults in the power system and predict the possible effects of faults on the power system.
- In terms of intelligent scheduling and control: digital twin technology can provide intelligent scheduling and management of the power system based on real-time data to optimize operational efficiency and energy saving. In this investigation, we use digital twin technology to optimize the load distribution of the power system and improve the energy utilization efficiency of the power system.

VII. CONCLUSION AND FUTURE WORK

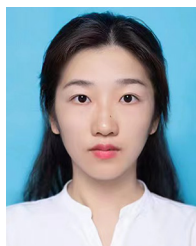
To solve the issue of the surge of data in the power grid model leading to a significant increase in system risk, we combined the Atrous Convolution method with the simulation of the digital twin power grid and adopted a new positive and negative sample selection method. Additionally, time series prediction and classification tasks are performed on ETT, electricity, and UCR datasets. Experimental results show that the algorithm proposed reduces the error rate by 4.38% and 5.21% in the prediction task and improves the accuracy by 8.39% and 12.73% in the classification task, which can forecast the power grid operation, and warns of potentially dangerous situations, assisting in ensuring the stable operation of the power grid system.

The key issue of the proposed prediction method is to forecast the data set in its next steps. Anomaly detection and processing has always been a significant task of time series, though it has not been involved in this work. Currently, most models are trained offline using a large amount of historical data in the smart grids field, which may not be applicable to real-time online anomaly detection and also leads to the problem of lack of data in the initial stage of collecting data for real-time detection. In future directions, we will conduct in-depth research on anomaly detection methods.

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